



Asian Institute of Technology
Faculty of Science and Technology

AST02.21: E-business Development and Technology
August 2026 Semester

Assignment: E-Commerce Software/Platforms
Due 15 October 2026

E-commerce has transformed the way organizations conduct business. E-commerce software and platforms provide the technological infrastructure required to operate an online business. E-commerce platforms range from simple solutions designed for small businesses to sophisticated enterprise platforms that support millions of customers, multiple countries, complex product catalogs, and integration with enterprise systems. Examples include Shopify, WooCommerce, Adobe Commerce, Salesforce Commerce Cloud, and BigCommerce.

E-commerce businesses can choose from different types of e-commerce platforms depending on their business size, technical capabilities, budget, customization requirements, and growth strategy. There are three major platform types:

1. Open-Source E-Commerce Platform

(Software whose source code is available for customization. The business or developers usually manage hosting, customization, and maintenance.)

2. SaaS/Cloud-Based E-Commerce Platform

(A hosted platform provided as a service. The provider manages infrastructure, updates, security, and much of the technical operation.)

3. Enterprise E-Commerce Platform

(A sophisticated platform designed for large organizations with complex operations, high transaction volumes, multiple markets, and extensive integration requirements.)

I. Assignment Task

Select one platform from each of the following categories.

Type 1: Open-Source Platform

Choose **one** platform, such as:

- WooCommerce
- Adobe Commerce Open Source
- PrestaShop
- OpenCart

Type 2: SaaS/Cloud-Based Platform

Choose **one** platform, such as:

- Shopify
- BigCommerce
- Wix eCommerce

Type 3: Enterprise E-Commerce Platform

Choose **one** platform, such as:

- Salesforce Commerce Cloud
- Adobe Commerce
- SAP Commerce Cloud

II. Requirements for Each Platform

For each selected platform, prepare a detailed description covering the following areas.

A. Platform Introduction

Provide:

- Platform name
- Company/developer
- Year introduced
- Type of platform
- Target market
- Typical users
- Main industries served

Explain why the platform belongs to its particular category.

B. Platform Architecture

Describe how the platform works from a technical perspective.

Discuss:

- Hosting model

- Cloud/on-premises characteristics
- Database
- Application layer
- Web/storefront layer
- APIs
- Integration capabilities
- Scalability

Include a simple architecture diagram.

C. Major Features

Identify and explain at least **eight major features**.

Possible features include:

- Product catalog management
- Shopping cart
- Checkout
- Payment processing
- Order management
- Inventory management
- Customer management
- Promotion management
- Search
- Product recommendations
- Mobile commerce
- Analytics
- Marketing automation
- Multi-store support

D. Payment and Order Processing

Explain how the selected platform supports:

1. Customer checkout
2. Payment processing
3. Order confirmation
4. Order fulfillment
5. Shipping
6. Returns/refunds

E. Integration

Explain how the platform can integrate with other business systems.

Discuss possible integration with:

- ERP
- CRM
- Payment gateway
- Warehouse management system

- Inventory management system
- Logistics providers
- Marketing platforms
- Business intelligence/analytics systems
- Social media

Explain the importance of **APIs** and data integration.

F. Analytics and Reporting

Explain the analytics capabilities of the selected platform.

Discuss how the platform can help businesses analyze:

- Sales
- Customers
- Products
- Website traffic
- Conversion rates
- Shopping-cart abandonment
- Customer lifetime value
- Marketing campaigns

Give at least **three** examples of business decisions that could be supported by e-commerce analytics.

G. Security

Discuss the security capabilities and concerns associated with the platform.

Consider:

- User authentication
- Access control
- Data encryption
- Payment security
- Privacy
- Fraud prevention
- Security updates
- Backup and recovery

Explain why security is particularly important for e-commerce businesses.

H. Advantages and Disadvantages

Identify three advantages and three disadvantages of the selected platform.

III. Cross-Platform Comparison

After describing all three platforms, compare them using the following criteria.

Criteria	Open-Source Platform	SaaS Platform	Enterprise Platform
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Platform selected

Hosting

Initial cost

Operating cost

Ease of implementation

Customization

Scalability

Security management

Technical expertise

Integration

Analytics

Multi-store support

Best suited for

IV. Assignment Scenario

Case: Selecting an E-Commerce Platform for a Growing Retail Company
ABC is a medium-sized fashion retailer operating 15 physical stores. The company currently sells products through its stores and social media but does not have a fully integrated e-commerce website.

Management wants to establish an online store to support its digital transformation strategy.

The company has the following requirements:

- Approximately 20,000 products
- 15 physical stores
- One central warehouse
- Approximately 100,000 customers
- Online and offline inventory synchronization
- Credit-card and digital-wallet payments
- Mobile-friendly shopping

- Integration with its existing ERP system
- Customer loyalty program
- Product recommendations
- Sales and customer analytics
- Support for future expansion into neighboring countries
- Strong security
- Ability to handle significant increases in traffic during promotional campaigns

Please recommend one E-Commerce Platform for this case and justify your choice.

NOTE:

1. Please include all references in your report. If you use GenAI, please specify the one(s) you use.

2. Submit a report (10-15 pages) with a cover page, as a PDF file in TEAL. Please name your file as Ebiz_Assignment_your name